

Systems of Connected Particles

Part One

1. A block of mass 4 kg rests on a smooth (frictionless) table. On top of the block is a smaller block of mass 2 kg. I apply a horizontal force of 8 N to the lower block. The upper block does not slide or topple (so the blocks move together).
 - a. Draw a force diagram for the whole system. What is the acceleration of the blocks?
 - b. Draw a force diagram for the upper block. What horizontal force does the lower block exert on the upper block? What vertical force?
 - c. Draw a force diagram for the lower block and check that N2 holds horizontally and vertically.
 - d. Indicate clearly (perhaps using colours) any N3 force pairs in your three diagrams.
2. A box of mass 100 kg is in a lift of mass 900 kg, which is suspended from a light cable. Find the force of the cable on the lift, and the force of the lift on the box, when:
 - a. the lift is stationary.
 - b. the lift is accelerating upwards at 2 ms^{-2}
 - c. the lift is accelerating downwards at 5 ms^{-2} .
3. A man is standing still. He has a hat on his head.
 - a. Draw separate force diagrams for the system, the man, and the hat.
 - b. Explain (mentioning one of Newton's laws) why the weight of the hat is equal to the force of the man on the hat.
 - c. Explain (mentioning one of Newton's laws) why the force of the man on the hat is equal to the force of the hat on the man.
4. An 80 kg man is descending in a lift of mass 500 kg with an acceleration downwards of 1.5 ms^{-2} .
 - a. Draw diagrams showing the forces acting on:
 - i. the system;
 - ii. the man;
 - iii. the lift.
 - b. Use your diagrams in (i) and (ii) to find the tension in the lift cable, the force exerted by the floor on the man, and the force exerted by the man on the floor.
 - c. Check your diagram in part (iii) for consistency with N2.
5. A mass of 200 grams is suspended by a light vertical string. To a hook on the underside of the mass is attached a second light vertical string, which carries a mass of 300 grams. The masses are raised by a tension of 5 N in the upper string.
 - a. Find the acceleration with which they rise and the tension in the lower string.
 - b. If the upper string were cut and the masses were allowed to fall, what would be the tension in the other string?

Part Two

6. Two blocks, of masses 3 and 9 kg, rest on a smooth table. They are joined by a rod of negligible mass. I pull the lighter block in the direction away from the heavier block with a driving force of 36N.
 - a. Draw a force diagrams for:
 - i. the system
 - ii. the lighter block
 - iii. the heavier block
 - b. Calculate the acceleration and the tension in the rod.
 - c. What force acts on the heavier block to cause it to accelerate?

7. A car of mass 1000 kg is towing a trailer of mass 2000 kg along a smooth road at constant speed. The driving force is 4 kN, and the car is connected to the trailer by a light tow-bar.
 - a. Draw a force diagrams, including unknown resistance forces on both the car and the trailer, for:
 - i. the car
 - ii. the trailer
 - iii. the system
 - b. Calculate the size of the total resistance force on the system.
 - c. If the magnitude of the force of the bar on the car is 1 kN, find
 - i. the force of the bar on the trailer
 - ii. the resistance force on the trailer
 - iii. the resistance force on the car
 - d. Which of Newton's laws allows you to state that the force of the bar on the trailer is equal to the resistance force on the trailer?
 - e. The driver puts his foot down and the car starts to accelerate in the direction of travel at 1 ms^{-2} . Assuming that the resistance forces remain as calculated in part c, find:
 - i. the force of the bar on the trailer
 - ii. the force of the bar on the car
 - iii. the driving force
 - f. The driving force is increased to 28 kN. Assuming the resistance forces remain as before, find:
 - i. the resultant acceleration
 - ii. the force of the bar on the car.
8. Patience buys a trailer with mass 500 kg from a car dealer and attaches it to her car. When the car is in motion, there are constant resistances to the motion, of 600 N acting on the car and of 400N acting on the trailer.
 - a. Patience drives at a constant velocity. What is the tension in the tow-bar?
 - b. Could the tension in the tow bar ever be zero when the car and trailer are moving forward? Explain briefly.
9. An electric train consists of three coaches, their masses (starting from the front) being 40 tonnes, 24 tonnes, 36 tonnes. When the brakes are applied, only the wheels of the front coach are locked, and these produce a resistance of 100 kN. Find the retardation of the train and the force transmitted through each coupling. Are the couplings in tension or thrust (compression)?
10. An 80 kg woman sits in the driving seat of an open top sports car of mass 900 kg.
 - a. Find the force the car exerts on the road when stationary.

The car is now travelling at a constant speed. There is a constant resistance to motion of 600 N on the car and 100 N on the woman.

- b. Draw separate diagrams showing the forces acting on the woman and on the car.
- c. Find the force(s) exerted by the car seat on the woman. Also find the driving force.

The driving force is then increased to 5600 N

- d. Find the acceleration of the car and woman.
- e. Find the force(s) exerted by the car seat on the woman. What are the force(s) exerted on the car seat by the woman?

The car now has to suddenly brake in order to avoid hitting a child crossing the road. The car decelerates at 8 ms^{-2} .

- f. Find the horizontal force the seat belt exerts on the woman. Also find the braking force.
- g. If the car is trying to brake from 20 m s^{-1} (45 miles per hour) in 30 m, does she avoid hitting the child?